Høyskolen Kristiania

TEK306- Security in mobile applications

Course Description

Introduction

The course aims to give an introduction to mobile applications' vulnerability to attacks. What types of threats are excitating, and how can one hedge against them? The course deals with general threats, regardless of operating system, but focuses in particular on securing applications on the iOS and Android platforms.

Learning outcomes

Knowledge The student...

- has knowledge of the different types of threats that exist for mobile applications in general, and for applications on the iOS and Android platforms in particular
- has knowledge of the types of attacks that have been most dominant in recent times
- has sufficient knowledge of the structure and structure of the iOS and Android platforms related to possible threats
- is familiar with security models for iOS and Android

Skills The student...

- can identify possible vulnerabilities in mobile applications on the iOS and Android platforms
- can develop measures to secure mobile applications on the iOS and Android platforms

General competence The student...

- can account for the threat picture around mobile applications
- can explain how security can be incorporated as a natural element when developing mobile applications

The course is included in

Bachelor in Information Technology - Frontend and Mobile Development

Learning activities

The course consists of 12 lectures of about 2 hours and 12 teacher-led exercises of about 2 hours. The exercises are not mandatory, but in order to achieve the competence goals, students are expected to complete the exercises and also on their own put in extra effort where the exercises do not extend.

Recommended time spent

Participation in teaching and supervision - 48 hours Self-study - 110 hours

Independent preparation for presentation/discussion in class - 0 hours

Student work with projects, productions, assignments, etc. - 0 hours

Independent exercise / lab work / practical work individually or in groups - 0 hours

Completion and preparation of the exam - 42 hours

Total recommended time spent - 200 hours

Work tools

iOS and Android tools will be pointed out at course launch

Compulsory activity

Compulsory work requirements: Consists of 2 activities that together must be assessed approved

Individual qualification: G/IG (approved/not approved) Implementation: in groups

Can the work requirement be processed before final delivery? yes Verifiable: yes

Regulates access to: Written individual home exam

Coursework requirements are submitted or implemented in accordance with information provided by the course coordinator.

Coursework requirements are carried out within the duration of the course and must be registered as approved/not approved at least two weeks before the exam/exam period.

Approved coursework requirements qualify for admission to the exam. If the coursework requirements are not approved, the student will be notified of the exam.

Examination

Exam part: Written individual home exam Duration: 24 hours

Grading: National grading scale A - F (C not passed) Weighting: 100% of the full assessment

Aids: All aids allowed

Continuation

Continuation of coursework requirements: The student documents for qualification necessary during the re-sit weeks to take the exam.

Re-sit examination: 72 hours written individual home exam with new thesis text.

Teaching materials

In addition to literature and other teaching materials, scheduled teaching and other scheduled learning activities are always included in the current curriculum.

Comments

In order to succeed with the course, advanced programming knowledge is assumed on a par with the 2nd cohort at the Bachelor in IT or equivalent. Furthermore, the teaching requires that the students have experience in the development of mobile applications. Recommended prerequisites are: *Android programming*.